

Recap_m2_sp26

Questions from Ed

Probability language

I definitely struggle the most with the earliest portion of section 2. Being able to look at a word problem and write it out as " $P(A|B)$ " or " $P(A \cup B)$ " etc. all feels like gobbledygook in my brain. Review would be appreciated!

Probability language

General terms: Probability space, marginal probability, conditional probability, intersect, union, independence, disjoint

In a survey of US households, we find that 13.7% have experienced food insecurity.

Among households with food insecurity 38% have very low food security.

Household food insecurity affected 18.4 percent of households with children in 2024. In some of these food-insecure households only adults were food insecure, while in other households children also experienced food insecurity.

Probability language

60% of Americans receive coverage through employers

66% of Americans have private health insurance

Example problem setup

The Acme Corporation manufactures sticks of dynamite. Unfortunately (fortunately?), these sticks often fail in hilarious but harmless fashion.

Sticks manufactured at Plant A fail with a probability of 0.5

Wile E Coyote has a bag of 10 sticks of dynamite: 4 are from Plant A and 6 are from Plant B.

The Coyote pulls 1 stick from the bag at random and lights it.

The probability that Coyote picks a stick from Plant B and it fails is 0.2.

What is the probability that a stick from Plant B fails?

If a stick has failed, what is the probability that it is from plant A ?

Distributions syntax

Could you go over when to know which R functions to use:
`ppois()`, `dpois()`, `rpois()` for Poisson distributions?

Distributions

Distribution	Defined by:	Type of outcome	R notation
Normal	Mean and SD	Continuous	norm
Binomial	number and p	Binary (success or failure in n trials)	binom
Poisson	mean (λ)	Discrete count of events in an interval	pois

You should be familiar with what these distributions look like and what changes in the shape of the distribution as the key parameters change.

Which distribution?

What distribution would you think of for the following studies?

- ▶ Tracking the number of ER visits for car accidents during the weeks of winter
- ▶ Estimating the proportion of male and female children in a school who missed at least one day of school due to flu
- ▶ Estimating the number of minutes of exercise among students before and after new years day

SD vs SE

When do you use SD vs SE (σ vs $\sigma/\sqrt{n}$) when calculating the z scores and different probabilities? I noticed when we were calculating power, we would use SD for the distributions of H_a and H_o but then use SE for the z-scores and I was confused as to why.

Hypothesis testing and CIs

Could we spend some review time on hypothesis testing and confidence intervals, especially null distribution vs sampling distribution, standard error, two-sided p-values/shaded regions, using `pnorm()`, significance level vs p-value, and how confidence intervals connect to hypothesis tests?

Recap - confidence intervals and testing

How confidence intervals behave

Recall the form of a CI:

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

Where $z^* \frac{\sigma}{\sqrt{n}}$ is the **margin of error**.

How confidence intervals behave

The margin of error gets smaller when:

- ▶ z^* is smaller (i.e., you change to a smaller confidence level).
Thus, there is a trade-off between the confidence level and the margin of error.
- ▶ σ is smaller. You might be able to reduce σ if there is measurement error. Often times, the σ can't be reduced, it is just a characteristic of the population
- ▶ n is larger.

Interpretation of a CI

Not to be interpreted as a feeling- 95% confident- not correct!

Best interpretation in terms of confidence coefficients in relations to all possible samples: We calculated this interval using a method that gives an interval that includes the true population mean/proportion for 95 % of all possible samples (or long run performance).

NOTE: 5 out of 100 times, you will miss the true population mean/proportion (outside of the CI). You cannot say the probability of your sample is described by 95%, it describes the long run behavior, not just your sample. If you miss the mean, and it falls outside your CI range on that sample, then your probability is zero. CIs describe the long run behavior.

Interpretation of a CI

Interpretation of last statement: The 95 % confidence coefficient tells us that 95 % of all possible samples will give an interval that includes the true population value, but we do not know whether or not our particular study's results are one of the 95 % percent that do include the true population value or one of the 5 % that do not.

How hypothesis tests behave

- ▶ Statistical significance depends on sample size (since sample size determines the standard error of the sampling mean)
- ▶ Recall the form of the z-test:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{\text{magnitude of observed effect}}{\text{size of chance variation}} = \frac{\text{signal}}{\text{noise}}$$

- ▶ The numerator quantifies the distance between what you observe in the sample and the null hypothesized parameter.
- ▶ The denominator represents the size of chance variations from sample to sample

How hypothesis tests behave

- ▶ Statistical significance depends on:
 - ▶ The size of the observed effect ($\bar{x} - \mu$)
 - ▶ The variability of individuals in the population (σ)
 - ▶ The sample size (n)
 - ▶ Your criteria for rejection the null (α)

If you obtain a small p-value it is not necessarily because the effect size is large.

Interpretation of Hypothesis test result

If the null hypothesis is true, what fraction of results (repeated many times) would come out as extreme as the difference we found or more extreme by random chance?

$P=.003$. If the null hypothesis is 6.3 days is true, then less than 3 in 100 would give $\bar{x}$ than 5.4 or smaller.

Interpretation of Hypothesis test result

Could be making a type II error: incorrectly retaining a false null hypothesis.

Lack of significance, does not mean that the null hypothesis is actually true.

A small pvalue is not evidence of a big difference/effect

null distribution vs sampling distribution

two-sided p-values/shaded regions

significance level vs p-value

p-hacking

Could you also go over p-hacking and the some methods to reduce p-hacking?